15 When steam is passed over white hot coke, a mixture of combustible gases is obtained.

$$C(s) + H_2O(g) \underset{k_b}{\overset{k_f}{\rightleftharpoons}} H_2(g) + CO(g) \qquad \Delta H = +131 \text{ kJ mol}^{-1}$$

When equilibrium has been established, which of the following correctly describes what would happen if a proposed change is made to this system?

	proposed change	value of K_c	forward rate constant, k_f	backward rate constant, k_b
A	add catalyst	no change	increase	increase
B	add more C(s)	no change	increase	no change
C	increase volume	increase	decrease	decrease
D	increase temperature	increase	increase	decrease